

Radiation, Magnitudes & Spectra

Problem Set — Solutions

fizicyst

Section A — Objective

O1. Answer: (d). $L \propto R^2 T^4$; with R fixed and $T \rightarrow 2T$, L scales by $2^4 = 16$.

O2. Answer: (c). A cooler gas in front of a hot continuous source removes light at its own characteristic wavelengths, producing an absorption spectrum.

O3. Answer: (a). $m_1 - m_2 = -2.5 \log_{10}(f_1/f_2) = -2.5 \log_{10}(100) = -2.5(2) = -5$.

O4. Answer: (c) is false. Giants and supergiants are actually *cooler* than main-sequence stars of comparable luminosity — they're luminous because of their enormous size, not their temperature. (a), (b), and (d) are all correct.

O5. Answer: (c). $n = 3 \rightarrow n = 2$ is $H\alpha$, the first line of the Balmer series, at 656.3 nm — (a) is Lyman- α ($n=2 \rightarrow 1$), (b) is $H\beta$ ($n=4 \rightarrow 2$), (d) is Paschen- α ($n=4 \rightarrow 3$).

Section B — Subjective

S1.

$$(i) \frac{L}{L_{\odot}} = \left(\frac{R}{R_{\odot}}\right)^2 \left(\frac{T}{T_{\odot}}\right)^4 = (2.5)^2 \left(\frac{9500}{5775}\right)^4 \approx 6.25 \times 7.32 \approx 45.8.$$

$$(ii) M = M_{\odot} - 2.5 \log_{10}(L/L_{\odot}) = 4.83 - 2.5 \log_{10}(45.8) \approx 4.83 - 4.15 \approx 0.68.$$

$$(iii) m - M = 6.2 - 0.68 = 5.52 = 5 \log_{10}(d/10 \text{ pc}) \implies d = 10 \text{ pc} \times 10^{5.52/5} \approx 127 \text{ pc}.$$

S2.

$$(i) \tau \propto M/L \text{ and } L/L_{\odot} = (M/M_{\odot})^{3.5}, \text{ so}$$

$$\frac{\tau}{\tau_{\odot}} = \frac{M/M_{\odot}}{L/L_{\odot}} = \left(\frac{M}{M_{\odot}}\right)^{-2.5}.$$

$$(ii) \tau = 10 \text{ Gyr} \times (15)^{-2.5} \approx 10/871 \approx 0.0115 \text{ Gyr} \approx 11.5 \text{ Myr}.$$

(iii) Lifetime falls off steeply with mass ($\tau \propto M^{-2.5}$), so O/B stars burn through their fuel in only a few million years — orders of magnitude shorter than the age of the universe. Any O/B star observed today must therefore have formed very recently; the old ones have long since evolved off the main sequence.

S3.

- (i) $z = \frac{682.1 - 656.3}{656.3} \approx 0.0393$, so $v = zc \approx 0.0393 \times 3 \times 10^5 \approx 1.18 \times 10^4$ km/s (receding).
- (ii) Rest-frame peak: $\lambda_{\text{max,rest}} = \lambda_{\text{obs}}/(1+z) = 600/1.0393 \approx 577.4$ nm. Then $T = b/\lambda_{\text{max,rest}} = 2.898 \times 10^{-3}/577.4 \times 10^{-9} \approx 5.0 \times 10^3$ K.
- (iii) Redshift stretches every wavelength by the same factor $(1+z)$, so $\lambda_{\text{max,obs}} = (1+z)\lambda_{\text{max,rest}}$ — the curve's shape (hence "blackbody-ness") is preserved, only its horizontal scale changes, corresponding to a lower effective temperature via Wien's Law. This is exactly the division by $(1+z)$ used in (ii).